Hints: Number theory – 5

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Theorem 1: let g be the least positive element of $\{ax + by\}$. Division with remainder shows that g divides a and b, so $g = \gcd(a, b)$. If (x_0, y_0) and (x, y) are two solutions, then $a(x - x_0) = -b(y - y_0)$. Divide by $\gcd(a, b)$ and use coprimality to get the progression.

Hint 2. For $17x + 13y = 2026$: one particular solution is $(x, y) = (6, 148)$, so all solutions are $x = 6 + 13t$, $y = 148 - 17t$, $t \in \mathbb{Z}$.

Hint 3. McNugget: for every n there is a unique $y \in \{0, \dots, a-1\}$ with $by \equiv n \pmod{a}$. Show that n is representable with $x, y \geqslant 0$ if and only if $by \leqslant n$ for this y. The worst residue is $y = a - 1$, which gives the largest non-representable number $b(a-1) - a$.

Hint 4. Counting: for $0 \leqslant n \leqslant ab - a - b$, show that exactly one of n and $ab - a - b - n$ is representable. (If both were, adding the representations would represent $ab - a - b$.) So half of the $ab - a - b + 1$ numbers are non-representable, which gives $(a-1)(b-1)/2$.

Problem 2

Hint 1. Follow the proof sketched in Theorem 2. Dirichlet gives infinitely many (x, y) with $|x^2 - dy^2| < 2\sqrt{d} + 1$. So some value k occurs infinitely often, and among those solutions two have $x \equiv x'$, $y \equiv y' \pmod{k}$. Check that $(x + y\sqrt{d})/(x' + y'\sqrt{d})$ has integer coordinates and norm 1, and that it is not ± 1.

Hint 2. All solutions: let $\varepsilon = x_1 + y_1\sqrt{d}$ be the least solution > 1. If $u > 1$ is a solution with $\varepsilon^n \leqslant u < \varepsilon^{n+1}$, then $u\varepsilon^{-n}$ is a solution in $[1, \varepsilon)$, so it equals 1.

Hint 3. $\sqrt{13} = [3; \overline{1, 1, 1, 1, 6}]$ has odd period 5. The convergents are $3, 4, \frac{7}{2}, \frac{11}{3}, \frac{18}{5}$, and $18^2 - 13 \cdot 5^2 = -1$. Squaring gives the fundamental solution of the $+1$ equation: $(18 + 5\sqrt{13})^2 = 649 + 180\sqrt{13}$.

Hint 4. For $d = 34$ the fundamental solution of $x^2 - 34y^2 = 1$ is $35 + 6\sqrt{34}$. If the -1 equation were solvable, its least solution η would satisfy $\eta^2 = 35 + 6\sqrt{34}$, i.e. $x^2 + 34y^2 = 35$ and $xy = 3$, which is impossible. (Note that congruences cannot prove this: the -1 equation is solvable modulo every n.)

Problem 3

Hint 1. Reduce modulo a and modulo $a + 1$. You get $a \mid 2002$. Also $a + 1$ divides 2002 if n is even and 2000 if n is odd.

Hint 2. The only divisors $a > 1$ of $2002 = 2 \cdot 7 \cdot 11 \cdot 13$ with $a + 1 \mid 2000$ or $a + 1 \mid 2002$ are $a = 7$ (with n odd) and $a = 13$ (with n even). The case $a = 1$ is impossible.

Hint 3. For $a = 7$, look modulo 3: this forces n to be even, a contradiction. For $a = 13$ with n even, look modulo 8: $13^{n+1} \equiv 5$ while $2001 \equiv 1$, so $14^n \not\equiv 0 \pmod{8}$ and $n \leqslant 2$. The unique solution is $a = 13$, $n = 2$: $13^3 - 14^2 = 2001$.

Problem 4

Hint 1. Take a solution of $x^4 + y^4 = z^2$ with z minimal. Then (x^2, y^2, z) is a primitive Pythagorean triple, so (with y even) $x^2 = m^2 - n^2$, $y^2 = 2mn$, $z = m^2 + n^2$, where m, n are coprime and m is odd.

Hint 2. Then $x^2 + n^2 = m^2$ is again a primitive triple, so $n = 2rs$ and $m = r^2 + s^2$. Now $y^2 = 4rs(r^2 + s^2)$ with $r, s, r^2 + s^2$ pairwise coprime, so each is a square: $r = u^2$, $s = v^2$, $r^2 + s^2 = w^2$. This gives $u^4 + v^4 = w^2$ with $w < z$. A solution of $x^4 + y^4 = z^4$ would give one with z^2 in place of z.

Hint 3. Pythagorean triples: $(x, y, z) = k(m^2 - n^2,\, 2mn,\, m^2 + n^2)$ with $m > n > 0$ coprime and of opposite parity (and x, y possibly swapped). Derive this from $y^2 = (z - x)(z + x)$ in the primitive case.

Hint 4. For $x^2 + y^2 = 2z^2$, x and y have the same parity. Put $u = \frac{x+y}{2}$, $v = \frac{x-y}{2}$; then $u^2 + v^2 = z^2$. So the solutions are $x = u + v$, $y = u - v$ with (u, v, z) a Pythagorean triple, allowing $v \leqslant 0$. Example: $7^2 + 17^2 = 2 \cdot 13^2$.

Problem 5

Hint 1. First pin down x. An odd prime $p \mid x$ does not divide $xyz + 2$, so $p^3 \mid 2012 = 4 \cdot 503$, which is impossible. If $4 \mid x$, compare the powers of 2 on both sides. So $x \in \{1, 2\}$.

Hint 2. In both cases $y^3 + z^3 \equiv 0 \pmod{503}$. Since $3 \nmid 502$, cubing is a bijection mod 503, so $503 \mid y + z$. Write $y + z = 503k$ and divide by $y + z$, using $y^2 - yz + z^2 = (y + z)^2 - 3yz$.

Hint 3. For $x = 2$ the equation becomes $y^3 + z^3 = 503(yz + 1)$, i.e. $k(y^2 - yz + z^2) = yz + 1$. Show that $k \geqslant 2$ is impossible. So $k = 1$ and $(y - z)^2 = 1$, giving $(x, y, z) = (2, 251, 252)$.

Hint 4. For $x = 1$ you get $yz = \frac{503^2 k^3 - 8}{3k+4}$. The bound $yz \leqslant (y + z)^2/4$ forces $k \leqslant 4$. Check $k = 1, 2, 3, 4$: only $k = 1$ gives an integer, and then $(y - z)^2 = 503^2 - 4yz$ is not a square. Answer: only $(2, 251, 252)$.